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## Research experience

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**Postdoctoral Researcher**, Max Planck Institute for Gravitational Physics Nov 2022 –

- Used numerical data to perform resummations of weak-field expressions for the radiated energy [1] and scattering angle [4] of a point-particle source, extending their validity into the strong-field regime.
- Extended the Spectral Einstein Code (SPEC) to be capable of producing the most accurate simulations of hyperbolic binary black hole encounters to date [9, 3], some of which have been made publically available [10]
- Produced the first numerical relativity simulations of highly asymmetric mass-ratio hyperbolic binary black hole encounters (mass ratio 1:10) and performed the first measurements of disparate scattering angles due to asymmetric gravitational wave emission [3].
- Used numerical relativity scattering simulations to extract high-order self-force coefficients and cross-validate post-Minkowskian calculations and effective-one-body models [2, 3].

**Postdoctoral Research Fellow**, University of Southampton Jun 2022 – Oct 2022

- Used numerical data to extract high-order weak-field expansions of the scattering angle due to a point-particle source [5].

**Ph.D. Student**, University of Southampton Sep 2018 – Apr 2022

- Developed and implemented a numerical method for calculating perturbations due to a point-particle source in black hole perturbation theory [15, 16].
- Performed the first-ever calculation of the post-geodesic correction to the scattering angle of a point-particle in a black hole spacetime [15].
- Derived analytic formulae for scattering geodesics in a black hole spacetime [16] and implemented them in the KerrGeodesics package of the Black Hole Perturbation Toolkit.

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## Education

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**Ph.D. in Mathematical Sciences**, University of Southampton Sep 2018 – Apr 2022

- Thesis title: Self-force in hyperbolic black hole encounters.
- Advisor: Prof. Leor Barack

**MPhys in Physics**, The University of Manchester Sep 2014 – Jun 2018

- Project title: Constraints on the neutrino sector using current and future cosmological data.
- Advisor: Dr Eleonora Di Valentino

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## Prizes and funding awards

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**Government of Ireland Postdoctoral Fellowship**, Research Ireland. 2026

- Funding for a two year tenure within the Relativity Group at University College Dublin (UCD), Ireland.
- Won as part of the Government of Ireland Postdoctoral Fellowship Programme 2025.

**Junior Investigator Fellowship**, Fundação para a Ciência e a Tecnologia (FCT). 2025

- Funding for a three year tenure at the Center for Astrophysics and Gravitation (CENTRA) within the Instituto Superior Técnico (IST), Portugal.
- Won as part of the 7th Call for the Stimulus of Scientific Employment, Individual Support (CEECIND).

**Seal of Excellence**, European Commission Horizon Europe. 2025 & 2026

- Gravitational Waves from Hyperbolic Encounters (GWHypE) is recognised as a highly rated proposal for the Marie Skłodowska-Curie Actions Postdoctoral Fellowships 2024 (86%) and 2025 (96%).

**Doctoral Prize Fellowship**, Engineering and Physical Sciences Research Council (EPSRC). 2022

- Funding for the Postdoctoral Research Fellow position at the University of Southampton.

**Best Student Talk Runner Up**, 25th Capra Meeting on Radiation Reaction in GR. Jun 2022

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## Featured publications

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- [1] L. Barack, R. Gonzo, B. Leather, **O. Long**, and N. Warburton, “Resummed energy loss in extreme-mass-ratio scattering using critical orbits,” *Phys. Rev. D*, vol. 113, no. 10, p. 104042, 2026. DOI: 10.1103/pzzz-dl2b. arXiv: 2602.10089.
- [2] **O. Long**, H. P. Pfeiffer, L. E. Kidder, and M. A. Scheel, “Black-hole scattering with numerical relativity: Self-force extraction and post-Minkowskian validation,” *Phys. Rev. D*, vol. 112, no. 12, p. 124038, 2025. DOI: 10.1103/bdsb-sp9c. arXiv: 2511.10196.
- [3] **O. Long**, H. P. Pfeiffer, A. Buonanno, G. U. Jakobsen, G. Mogull, A. Ramos-Buades, H. R. Rüter, L. E. Kidder, and M. A. Scheel, “Highly accurate simulations of asymmetric black-hole scattering and cross validation of effective-one-body models,” *Phys. Rev. D*, vol. 112, no. 12, p. 124039, 2025. DOI: 10.1103/p6fx-7798. arXiv: 2507.08071.
- [4] **O. Long**, C. Whittall, and L. Barack, “Black hole scattering near the transition to plunge: Self-force and resummation of post-Minkowskian theory,” *Phys. Rev. D*, vol. 110, no. 4, p. 044039, Aug. 2024. DOI: 10.1103/PhysRevD.110.044039. arXiv: 2406.08363.
- [5] L. Barack, Z. Bern, E. Herrmann, **O. Long**, J. Parra-Martinez, R. Roiban, M. S. Ruf, C.-H. Shen, M. P. Solon, F. Teng, and M. Zeng, “Comparison of post-Minkowskian and self-force expansions: Scattering in a scalar charge toy model,” *Phys. Rev. D*, vol. 108, no. 2, p. 024025, Jul. 2023, **Co-lead author** (follows HEP alphabetical author list convention). DOI: 10.1103/PhysRevD.108.024025. arXiv: 2304.09200.

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## Other publications

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- [6] A. Vaswani, L. Barack, **O. Long**, and R. Panosso Macedo, “Time-domain framework for the Teukolsky equation with a particle source using comoving hyperboloidal coordinates,” Jun. 2026. arXiv: 2606.26339.
- [7] A. Gamboa *et al.*, “Accurate waveforms for generic planar-orbit binary black holes: The multipolar effective-one-body model SEOBNRv6EHM,” May 2026. arXiv: 2605.28715.
- [8] C. Whittall, L. Barack, and **O. Long**, “Frequency-domain self-force calculations using Gegenbauer reconstruction,” *Phys. Rev. D*, vol. 112, no. 12, p. 124045, 2025. DOI: 10.1103/9qsl-pr63. arXiv: 2509.19439.
- [9] I. B. Mendes, N. L. Vu, **O. Long**, H. P. Pfeiffer, and R. Owen, “Parameter control for binary black hole initial data,” *Phys. Rev. D*, vol. 112, no. 12, p. 124049, 2025. DOI: 10.1103/zh31-bbtm. arXiv: 2509.07291.
- [10] M. A. Scheel *et al.*, “The SXS Collaboration’s third catalog of binary black hole simulations,” *Classical and Quantum Gravity*, vol. 42, no. 19, p. 195017, Oct. 2025. DOI: 10.1088/1361-6382/adfd34. arXiv: 2505.13378.
- [11] A. Abac *et al.*, “The Science of the Einstein Telescope,” *Journal of Cosmology and Astroparticle Physics*, vol. 2026, no. 03, p. 081, Mar. 2026. DOI: 10.1088/1475-7516/2026/03/081. arXiv: 2503.12263.
- [12] L. J. G. Da Silva, R. Panosso Macedo, J. E. Thompson, J. A. V. Kroon, L. Durkan, and **O. Long**, “Hyperboloidal discontinuous time-symmetric numerical algorithm with higher order jumps for gravitational self-force computations in the time domain,” Jun. 2023. arXiv: 2306.13153.
- [13] N. Afshordi *et al.*, “Waveform modelling for the Laser Interferometer Space Antenna,” *Living Rev. Rel.*, vol. 28, no. 1, p. 9, Oct. 2025. DOI: 10.1007/s41114-025-00056-1. arXiv: 2311.01300.
- [14] M. Boschini *et al.*, “Extending black-hole remnant surrogate models to extreme mass ratios,” *Phys. Rev. D*, vol. 108, no. 8, p. 084015, Oct. 2023. DOI: 10.1103/PhysRevD.108.084015. arXiv: 2307.03435.
- [15] L. Barack and **O. Long**, “Self-force correction to the deflection angle in black-hole scattering: A scalar charge toy model,” *Phys. Rev. D*, vol. 106, no. 10, p. 104031, Nov. 2022, **Co-lead author**. DOI: 10.1103/PhysRevD.106.104031. arXiv: 2209.03740.
- [16] **O. Long** and L. Barack, “Time-domain metric reconstruction for hyperbolic scattering,” *Phys. Rev. D*, vol. 104, no. 2, p. 024014, Jul. 2021. DOI: 10.1103/PhysRevD.104.024014. arXiv: 2105.05630.

## Invited seminars (total: 4)

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- CENTRA Seminar, Instituto Superior Técnico, “Black hole scattering with numerical methods and the resummation of weak-field analytic expressions”, 30th April 2026.
- WQFT Seminar, Humboldt University of Berlin, “Black hole scattering in the strong-field regime: Merging post-Minkowskian theory with numerical methods”, 25th November 2024.
- Gravity Seminar, University of Southampton, “Black hole scattering in the strong-field regime: Merging post-Minkowskian theory with numerical methods”, 7th November 2024.
- Séminaire Amplitudes et Gravitation sur l’Yvette, Institut des Hautes Études Scientifiques, “Black hole scattering in the strong-field regime: Merging post-Minkowskian theory with numerical methods”, 16th October 2024.

## Invited talks (total: 12)

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- Amplitudes, Strong-Field Gravity and Resummation, The Nordic Institute for Theoretical Physics, “Black hole scattering in the strong-field regime: Merging post-Minkowskian theory with numerical methods”, 15th April 2026.
- Milestones in Amplitudes for Gravitational Waves and Cosmology, ETH Zürich, “Black hole scattering in the strong-field regime: Merging post-Minkowskian theory with numerical methods”, 25th March 2026.
- Hyperboloidal Foliation and their Application, Erwin Schrödinger Institute, Universität Wien, “Hyping up self-force scattering: Methods and applications in the pre-hyperboloidal era”, 13th January 2026.
- GWSky kick-off meeting, Max Planck Institute for Gravitational Physics, “NR, PM, PN, GSF comparisons and transferring information from one method to another”, 9th December 2025 (joint with Raj Patil).
- 2nd Annual Workshop on Self-Force and Amplitudes, University of Southampton, “Applications of numerical self-force scattering simulations”, 10th September 2025.
- EOB@Work25, Istituto Nazionale di Fisica Nucleare, “Highly accurate simulations of asymmetric black-hole scattering and cross validation of effective-one-body models”, 4th September 2025.
- Mathematical Methods for the General Relativistic Two-body Problem, National University of Singapore, “Putting the hype in hyperbolic black hole scattering”, 14th August 2025.
- Fundamental Physics Meets Waveforms With LISA, Max Planck Institute for Gravitational Physics, “Modelling of unbound binary black hole encounters”, 6th September 2024.
- Gravitational Self-Force and Scattering Amplitudes Workshop, The Higgs Centre for Theoretical Physics, “Comparing numeric and analytic methods for black hole scattering in unequal mass systems”, 20th March 2024.
- Gravitational Waves meet Amplitudes in the Southern Hemisphere, International Center for Theoretical Physics South American Institute for Fundamental Research, “Self-force meets post-Minkowskian in the scattering regime”, 24th August 2023.
- Infinity on a Gridshell, Niels Bohr Institute, “Hyperbolic self-force calculations within a hyperboloidal framework”, 10th July 2023.
- QCD meets Gravity 2022, Universität Zürich, “Extraction of high-order post-Minkowskian results from scattering self-force calculations”, 13th December 2022.

## Contributed talks (total: 15)

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- 29th Capra Meeting on Radiation Reaction in General Relativity, Université Libre de Bruxelles, “Resummed radiative observables in eccentric and unbound extreme-mass-ratio systems using critical orbits”, 2nd July 2026.
- New Frontiers of Numerical Relativity, Universitat de les Illes Balears, “Highly accurate simulations of asymmetric black-hole scattering and cross validation of effective-one-body models”, 22nd July 2025.

- 24th International Conference on General Relativity and Gravitation and 16th Edoardo Amaldi Conference on Gravitational Waves, Glasgow, “Highly accurate simulations of asymmetric black-hole scattering and cross validation of effective-one-body models”, 15th July 2025.
- “Hyperbolic Binary Black Hole encounters with the Spectral Einstein Code”, NR Community Call, 2nd December 2024.
- Simulating Extreme Spacetimes with **SpEC** and **SpECTRE**, The Institute for Computational and Experimental Research in Mathematics, Brown University, “Hyperbolic Binary Black Hole encounters with **SpEC**”, 7th August 2024.
- 27th Capra Meeting on Radiation Reaction in General Relativity, National University of Singapore, “Double the hype: Hyperboloidal framework for self-force in hyperbolic black hole encounters”, 17th June 2024.
- 26th Capra Meeting on Radiation Reaction in General Relativity, Niels Bohr Institute, “Comparison of post-Minkowskian and self-force expansions: Scattering in a scalar charge toy model”, 4th July 2023.
- LISA Symposium XIV, “Self-force in hyperbolic black hole encounters”, 25th – 29th July 2022. 
- 23rd International Conference on General Relativity and Gravitation, Chinese Academy of Science via Zoom, “Self-force in hyperbolic black hole encounters”, 5th July 2022. 
- 25th Capra Meeting on Radiation Reaction in General Relativity, University College Dublin, “Self-force in hyperbolic black hole encounters”, 22nd June 2022. 
- BritGrav22, University of Glasgow via Zoom, “Self-force in hyperbolic binary-black-hole encounters”, 4th April 2022. 
- 24th Capra Meeting on Radiation Reaction in General Relativity, Perimeter Institute via Zoom, “Time-domain metric reconstruction for hyperbolic scattering”, 10th June 2021. 
- BritGrav21, University College Dublin via Zoom, “Towards a self-force calculation of the scatter angle in hyperbolic encounters”, 13th April 2021. 
- LISA Symposium XIII, “Towards a self-force calculation of the scatter angle in hyperbolic encounters”, 1st – 3rd October 2020. 
- 23rd Capra Meeting on Radiation Reaction in General Relativity, University of Texas at Austin via Zoom, “Towards a self-force calculation of the scatter angle in hyperbolic encounters”, 24th June 2020. 

### Posters (total: 1)

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- 3rd meeting of the GWVerse COST action, Institute for Fundamental Physics of the Universe, International School for Advanced Studies, “Time-domain metric reconstruction using the Hertz potential”, 13th – 16th January 2020.

### Other events attended (total: 11)

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- In Pursuit of Gravitational Waves: Solving the Two-Body Problem in General Relativity, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), 20th – 22nd October 2025.
- Black Hole Perturbation Toolkit Workshop (via Zoom), The Institute for Computational and Experimental Research in Mathematics, Brown University, 25th – 27th July 2022.
- From Scattering Amplitudes to Gravitational-Wave Predictions for Compact Binaries, Universität Zürich & ETH Zürich, 4th – 15th July 2022.
- Advances and Challenges in Computational Relativity Workshop (Online), The Institute for Computational and Experimental Research in Mathematics, Brown University, 14th – 18th September 2020.
- Black Hole Perturbation Toolkit Workshop (Online), Astronomical Institute of the Academy of Sciences of the Czech Republic, 25th – 27th May 2020.
- Kavli RISE Summer School on Gravitational Waves, University of Cambridge, 23rd – 27th September 2019.

- 22nd International Conference on General Relativity and Gravitation and 13th Edoardo Amaldi Conference on Gravitational Waves, Palau de congressos de Valencia, 8th – 12th July 2019.
- 22nd Capra Meeting on Radiation Reaction in General Relativity, Centro Brasileiro de Pesquisas Físicas, 17th – 21st June 2019.
- LISA Waveform Working Group Meeting, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), 13th – 15th May 2019.
- BritGrav19, Durham University, 15th – 16th April 2019.
- Black Hole Perturbation Toolkit Workshop, University College Dublin, 19th – 21st March 2019.

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### Mentorship and supervision

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Francisco Grippa

Nov 2025 – Apr 2026

- Visiting PhD student from Salerno University, co-supervised with Harald Pfeiffer.
- Project: Initial data for hyperbolic black hole encounters using the Extended Conformal Thin Sandwich (XCTS) approach.

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### Other community service

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- Member of the Simulating eXtreme Spacetimes (SXS) Social Media Team. Aug 2024 –
- Member of the Capra community’s Equality, Diversity and Inclusion (EDI) Team. Jun 2024 –
- Reviewer for *Physical Review Letters*, *Physical Review D*, and *SciPost Physics*. Mar 2023 –
- Discussion chair for “Scattering – What? Why? How?”, Fundamental Physics Meets Waveforms With LISA, Max Planck Institute for Gravitational Physics, 6th September 2024.
- Discussion chair for “Self-force meets Amplitudes”, 26th Capra Meeting on Radiation Reaction in General Relativity, Niels Bohr Institute, 4th July 2023.

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### Research collaborations

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#### Simulating eXtreme Spacetimes (SXS) collaboration

Nov 2022 –

- Contributor to the Spectral Einstein Code (SPEC).
- Member of the SXS Social Media Team.

#### The Black Hole Perturbation Toolkit

Mar 2019 –

- Contributor to the KerrGeodesics package.

#### Laser Interferometer Space Antenna (LISA) Consortium

Oct 2018 –

- Lead of the EMRI subproject of the Waveform Parameter Space for Astrophysical Sources project.
- Member of the Waveform Working Group and LISA Early Career Scientists (LECS).

#### LIGO Scientific Collaboration (LSC)

Apr 2023 – Nov 2023

- Member of the Waveform Working Group.

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### Computing experience

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- Extensive experience with high-performance computing (HPC) on clusters.
- Extensive experience with running and troubleshooting binary black hole simulations with SPEC.

**Advanced:** C/C++, Python, Mathematica, Git, Linux, MacOS,  $\text{\LaTeX}$ .

**Intermediate:** Bash, OpenMP, Slurm, Paraview, Windows.

**Some experience:** Perl, Cython, OpenMPI.

## Teaching experience

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**Postgraduate Student Demonstrator**, University of Southampton

Sep 2018 – Jan 2022

- Numerical Methods in Python: Run workshops based on problem sheets and coursework.
- Maths for Physicists: Run classes demonstrating mathematical techniques and coursework marking.
- Maths for Engineers: Run drop-in workshops, marking of exams, and exam writing.